Marmara Üniversitesi. Electrical and Electronics Engineering Department
EE 3011: Electronics I (Spring 2025)

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Course Information	Text: Sedra/Smith, *Microelectronic Circuits 5th Ed.*	Grading	Course Work 15%	Lab Work 15%	Mid-term Exam 30%	Final Exam 40%

Week	Events	Topics to cover	Ch	Sec	Lab Activity	Exercises
1 Feb 17-21	---	**Introduction to Physics of Semiconductors.** Conduction in Semiconductors, N-type and P-type Semiconductors, The PN junction.	3	7	NL	(**3**) 112,113,114,116,118.
2 Feb 24-28	---	**Diodes**. Introduction, The Ideal diode, Terminal characteristics of the Diode.	3	1-2	Experiment (1) LTSpice	(**3**) 2,3,4,6,9,10.
3 Mar 3-7	Course drop/add Week	**Diode Circuit Analysis**. Diode Models, Graphical and Analytical Diode Circuit Analysis.	3	3	Experiment (2) LTSpice	(**3**) 17,20,22,25,26, 40,41,46,47.
4 Mar 10-14	---	**Diode Applications**. The Zener diode. Half and Full-wave Rectifiers, Limiting and Clamping Circuits, Voltage Doublers. Other Diode types.	3	4-6, 8	Experiment (3) Diode ckt Simulation	(**3**) 64,65,66,71,76,78, 82,93,98,105.
5 Mar 17-21	---	**MOS Field-Effect Transistors (MOSFETs).** Device Structure and Modes of Operation. Types, Symbols and Conventions, Current-voltage (IV) Characteristics.	4	1-2	Experiment (4) Basic Diode Circuits	(**4**) 4,6,12,15,19,23.
6 Mar 24-28	Quiz 1 March 25 (In-class)	**MOSFET DC Circuits.** Biasing and Circuit DC Analysis.	4	3	Experiment (5) Diode Rectifier Circuit	(**4**) 35,36,37,41,42,44,47.
7 Apr 2-4	Mar 29 - Apr 1 Holidays	**---**	---	---	NL	---
8 Apr 7-11	---	**MOSFET Amplifiers.** Small Signal Operation, Small-signal Equivalent Circuits. Single-stage Amplifiers (CS, CG and CD).	4	4-6	Experiment (6) MOSFET DC Circuits	(**4**) 63,69,74,75,77.
April 14-20	**Mid-term Exams period**					
10 Apr 21-25	---	**MOSFET Amplifiers.** Single-stage Amplifiers (CS, CG and CD).	4	7	NL	(**4**) 81,85,88.
11 Apr 28-May 2	Labor Day: May 1	**Bipolar Junction Transistors (BJTs).** Structure and Modes of Operation, Types, Symbols and Conventions, Current-voltage (IV) Characteristics.	5	1-2	Experiment (7) MOSFET Amplifiers	
12 May 5-9	---	**BJT DC Circuits**. Biasing. BJT as a switch.	5	4	NL	(**5**) 3,4,7,13,20,31,39.
13 May 12-16	Midterm Make-up (Excused)	**BJT Amplifiers.** Small Signal Operation, Small-signal Equivalent Circuits. Single-stage Amplifiers (CE, CB and CC).	5	6	Experiment (8) BJT Characterization	(**5**) 69,71,72,74,79,82, 84,86.
14 May 19-23	Quiz 2 May 20 (In-class)	**BJT Amplifiers.** Single-stage Amplifiers (CE, CB and CC).	5	7	NL	(**5**) 104,106,107,115, 116,124.
15 May 26-30	---	**BJT Amplifiers.** Single-stage Amplifiers (CE, CB and CC).	5	7	Experiment (9) BJT Amplifiers	(**5**) 130,131,134.
16 June 2-4	June 5-9 Holidays	**CMOS Logic Circuits.** Overview, the CMOS Inverter, Other CMOS Logic circuits, CMOS transistor sizing	10	1-3	---	(**10**) 18,25,28.
17 June 10-13	Project submission Week	**Review**	---	---	---	---
June 16-29	**Final Exams period**	**June 30 - July 13 (Re-sit Final Exams period)**				